

COMPENG 2DX3 - Microprocessor System Project

Final Project Report

Instructors: Dr. Elamien, Dr. Athar, Dr. Haddara

Due Date: Tuesday, Apr. 7, 2026

Aundre Surendra – L05 – surena10 - 400559794

As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario. Submitted by [**Aundre Surendra, 400559794**]

Table of Contents

Design Overview	3
Features.....	3
MSP-EXP432E401Y Microcontroller.....	3
28BYJ-48 Stepper Motor.....	4
ULN2003 Driver Board.....	4
VL53L1X Time of Flight Sensor	4
Serial Communication	4
Programming Language	4
General Description.....	5
Block Diagram	6
Device Characteristics	7
MSP432E401Y Microcontroller	7
28BYJ-48 Stepper Motor + ULN2003 Driver Board	7
VL53L1X Time of Flight Sensor	7
Detailed Description.....	8
Distance Measurement	8
Visualization	9
Application Example.....	10
Instructions	10
Scanned Location	11
Location Images	12
Expected Output.....	13
Limitations.....	15
Floating Point and Trigonometric Limitations.....	15
Maximum Standard Serial Communication Rate	15
Communication Methods & Speed	16
Primary Limitation on Speed.....	16
Circuit Schematic.....	17
Programming Logic Flowchart.....	18

Design Overview

This device is an integrated LiDAR (Light Detection and Ranging) system, created using smaller, inexpensive components. The system is driven by a Texas Instrument MSP-EXP432E401Y microcontroller, an onboard button, 3 onboard LEDs, a ULN2003 driver board, a 28BYJ-48 stepper motor and a VL53L1X Time of Flight (TOF) sensor mounted on the stepper motor.

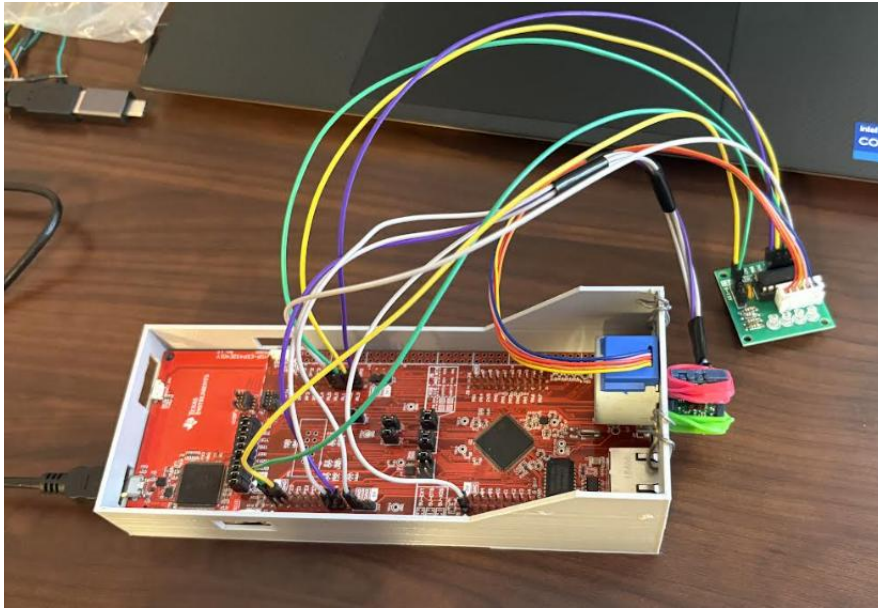


Figure 1. Image of LiDAR System

Features

This section holds the performance features of each hardware element used in this system.

MSP-EXP432E401Y Microcontroller

- ARM Cortex-M4 32-bit CPU with floating-point support
- System Bus Speed: 22 MHz
- Max Bus Speed: 120 MHz
- 3 LED status indicators
- 1 Onboard Button for user to control system
- 1 MB flash memory
- 256 kB SRAM
- Cost: \$47.99 – \$54.49

28BYJ-48 Stepper Motor

- Unipolar 4-phase stepper motor
- 11.25° step angle
 - 32 steps per rotation
- Operating Voltage: 5V – 12V
- Cost: \$5.00 – \$7.49

ULN2003 Driver Board

- 4 onboard LEDs to indicate 4-step stepper motor sequence
- Cost: \$1.50 – \$2.50

VL53L1X Time of Flight Sensor

- Max Accurate Distance Measurement: 4m
- Operating Voltage: 2.6V – 5.5V
- Emits 940 nm infrared laser
- Cost: \$12.00 – \$30.00

Serial Communication

- I2C communication between TOF and microcontroller at 100 kbps
- UART communication between microcontroller and PC configured to 115200 bps

Programming Language

- MATLAB used for serial communication between microcontroller and PC
 - Serial data acquisition using serialport interface
 - Performs trigonometric coordinate transformation
 - Creates 3D plot using mesh function

General Description

This embedded spatial measurement system makes use of a VL53L1X Time of Flight sensor to acquire information about the area around a user, simulating a portable LiDAR for the purpose of indoor use. This device is built on the basis of the MSP432E401Y microcontroller, serving as the primary hub for data acquisition, signal processing, and mechanical control. By combining digital logic communication with precise motor steps, the system can generate a 2D map of its surroundings with great accuracy.

The system initiates signal acquisition through the use of a 940nm infrared laser emitted by the TOF sensor, along with a single photon avalanche diode (SPAD) array to detect reflected light. The sensor measures the time interval over which the emitted pulses of light reach the nearest object and reflect back to the detector. Based on the known speed of light as well as the time it takes to detect the reflection, the distance can be calculated. Within the sensor, the analog signals detected undergo preprocessing to filter out any noise which is then converted to a 16-bit digital distance value via the onboard analog-to-digital (ADC) converter. The digital data is stored on the TOF sensor to be transmitted back to the microcontroller for use.

This system operates on the “observe, reason, act” cycle. The process is triggered by a hardware interrupt when the user presses the onboard microcontroller button: PJ0. When pressed, the onboard LED PF4 is turned on during the falling edge transition, indicating the button has been pressed. Then the measurement process begins, with the TOF scanning a distance, sending the data to the board via I2C communication which is then sent to the PC via UART communication. Each time a distance is measured, the PN1 LED flashes, indicating the distance measurement is successful. Right before the data is sent to the PC via UART, the PN0 LED flashes, indicating the transmission process has begun.

Following each transmission, the 28BYJ-48 stepper motor executes the act phase, rotating 11.25 degrees clockwise (32 steps per revolution) via a 4-step sequence on GPIO Port L. This process repeats until the motor has spun 360 degrees, at which point the motor rotates 360 degrees counterclockwise to its original position.

The distance measurements which were transmitted through the UART to the PC are visualized in 3D through the use of MATLAB. The visualization program in MATLAB converts the distance measurements (d) to their cartesian coordinates using trigonometric functions. This is done by applying cartesian conversion formulas to create a 2D layer of the measured area for every 360-degree rotation. The resulting layers are then stacked on each other by using the distance travelled by the sensor in the z direction each cycle. Using the z displacement, a 3D point cloud of the area is generated, reconstructing a layout of the area.

Block Diagram

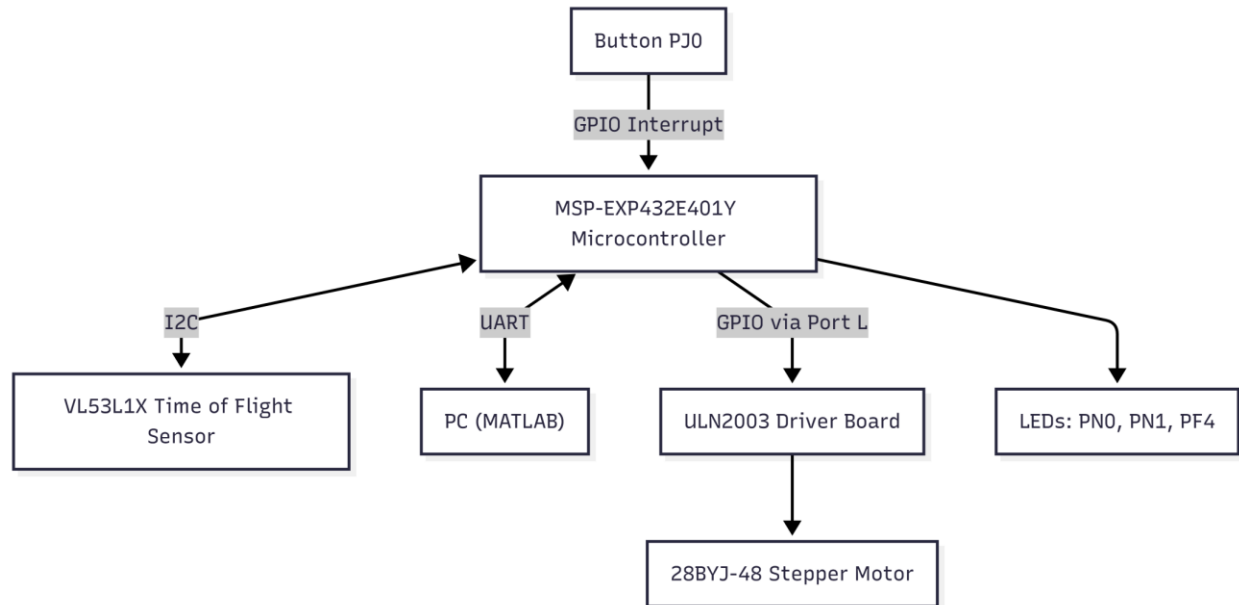


Figure 2. Block Diagram of LiDAR System

Device Characteristics

MSP432E401Y Microcontroller

Parameter	Value
Clock Speed	22 MHz
Button Interrupt Status LED	PF4
Distance Measurement Status LED	PN1
UART Status LED	PN0
Onboard Button	PJ0
UART Baud Rate	115200 bps
Serial Port	COM3



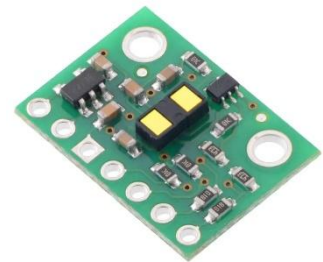
28BYJ-48 Stepper Motor + ULN2003 Driver Board

Board Pin	Microcontroller Pin
IN1	PL0
IN2	PL1
IN3	PL2
IN4	PL3
+5V	5V
-5V	GND



VL53L1X Time of Flight Sensor

TOF Pin	Microcontroller Pin
Vin	3.3V
GND	GND
SDA	PB3
SCL	PB2



Detailed Description

Distance Measurement

To capture distance measurements, a VL53L1X Time of Flight distance sensor was used by integrating a 940 nm infrared laser and a single photon avalanche diode (SPAD) array. This sensor measures the absolute time it takes for a photon to travel to a target and reflect back to the SPAD array (Δt). Since the speed of light is constant (c), the distance can be calculated in millimeters using the following equation:

$$d = \frac{\Delta t \times c}{2}$$

Since the time includes the distance traveled both ways, the total is divided by 2 to get the distance traveled by the photon in one direction.

Before the acquisition of data starts, there are some actions which are required first to synchronize the system components. The MSP-EXP432E401Y microcontroller first initializes the PLL, system clock, GPIO ports, I2C, UART, and status LEDs. The system then waits for an interrupt, remaining idle until the PJ0 onboard button has been pressed. This interrupt toggles the spin variable, triggering the loop which allows the TOF sensor to begin scanning a distance.

Once the distance is calculated, it is sent to the microcontroller using I2C communication, where the microcontroller is the leader and the sensor is the follower. The communication occurs over a serial, 8-bit oriented, bidirectional wire transfer where a serial clock line (SCL) sets the speed of the transfer at 100 kbit/s and the serial data line (SDA) carries the 16-bit distance value in two 8-bit packets – Most Significant Byte (MSB) and Least Significant Byte (LSB).

To achieve a high-resolution spatial map of the area, the system is programmed to take 32 measurements for every 360-degree rotation. This is because the angle of each step is a small value of 11.25 degrees, to capture more precise distance measurements. Using this, the number of measurements per rotation is derived from the following relationship:

$$\frac{360^\circ}{11.25^\circ} = 32 \text{ steps per rotation}$$

Once the scan has been complete, the motor rotates 360 degrees counter-clockwise back to its original position to repeat the process again until the PJ0 button has been pressed again. If the PJ0 button is pressed again, if a measurement is taking place, it will finish before rotating back to the homing position and stopping. The system then enters the visualization stage where a 3D plot of the area measured is created.

Visualization

The distance data sent to the microcontroller from the TOF is visualized using MATLAB, creating a 3D mesh representation of the area measured.

The visualization begins by establishing a serial connection with the microcontroller using the `serialport()` function. The serial communication is then configured by setting the baud rate to 115200 bps on the port COM3.

After receiving the start trigger, which is 's', a loop is run through 32 times (for the 32 steps per rotation). In each loop, the `sscanf` function is used to parse the UART data strings into an array containing the distance measurement (d), angle (θ), and range status. A check is done to ensure no initialization strings are used. The distance measurements are then converted into cartesian coordinates using the following equations:

$$x = d \times \cos\theta$$

$$y = d \times \sin\theta$$

$$z = \text{layer} \times z_increment$$

Where the $z_increment$ is the distance that the system moved along the z-axis (100mm), and the $layer$ is the index of the current cycle. By multiplying the layer number by the $z_increment$, a unique length coordinate is assigned to each set of 32 points measured. The z measurement allows the software to stack the 2D pieces of data on top of each other, creating a 3D spatial mapping.

An example of a measurement taken is at a distance of 300mm with an angle of 11.25° on layer 2:

$$x = 300mm \times \cos 11.25^\circ = 294.24mm$$

$$y = 300mm \times \sin 11.25^\circ = 58.53mm$$

$$z = 2 \times 100mm = 200mm$$

The resulting cartesian coordinate comes out to (294.24, 58.53, 200).

The final 3D visualization is generated using the `mesh()` and `reshape()` functions, which organize the coordinates into a grid and connect the data points to generate a mesh-like 3D model of the scanned area.

Application Example

An example of how this LiDAR system can be used to scan a 3D space is shown in the following section, along with detailed instructions. The example area tested is a hallway on the first floor of the Engineering Technology Building (ETB) located at McMaster University. Multiple figures have been provided below, providing a map as well as pictures of the location, and the final scanned figure. The instructions below can be followed as long as the required hardware and software – which can be found under the **Design Overview** section – are used, and the correct connections – which can be found under the **Circuit Schematic** section – are made.

Instructions

1. Connect the MSP432E401Y microcontroller to the computer using a micro USB cable.
2. Determine the COM port the devices are communicating through by opening windows device manager, checking under the “Ports” section, and noting the COM number beside the line labelled “XDS110 Class Application/User UART”.
3. Open the MATLAB script and adjust the port variable (port = “COM3”), which is on the second line, based on the COM number determined in step 2.
4. Make the correct connections between the MSP432E401Y microcontroller, VL53L1X time of flight sensor, and 28BYJ-48 stepper motor based on the connections located in **Circuit Schematic** section.
5. After ensuring the correct connections are made, load up the Keil code provided.
6. Click on the target settings in Keil to ensure they are correct by checking the following setting tabs:
 - a. “Target” Setting Tab
 - i. Arm compiler will use default compiler version 6
 - b. “Debug” Setting Tab
 - i. Ensure CMSIS-DAP Debugger is being used
 - ii. Click "Settings" to open the “CMSIS-DAP Cortex-M Target Driver Setup” window. Change the CMSIS-DAP – JTAG/SW Adapter to “XDS110 with CMSIS-DAP”
 - c. “C/C++ (AC6)” Setting Tab
 - i. Select –O0 as the Optimization level
7. After desired target settings set, click the following buttons in this order, located at the top left of Keil:
 - a. Translate
 - b. Build
 - c. Load
8. Click the reset button the on the microcontroller to initialize the program.

9. The Coordinate System: In the MATLAB code provided, the coordinate system is defined as follows to create an accurate spatial depiction of the area:
 - a. The X plane defines the horizontal width
 - b. The Y plane defines the vertical height
 - c. The Z plane defines the forward displacement between each measurement cycle
10. Execute the code in MATLAB, pressing enter when prompted to start the 3D mapping process.
11. Press the PJ0 button on the microcontroller to begin stepper motor rotation, capturing distance values at 11.25 degree intervals.
12. After every rotation, manually move the system along the Z-axis by 100mm.
13. When the desired number of layers is captured, press PJ0 again to finish data acquisition and execute the mesh() function, connecting the recorded data points to generate a 3D map of the scanned area.

Scanned Location

The image below is a map of the Engineering Technology Building (ETB), with the scanned hallway highlighted in yellow.

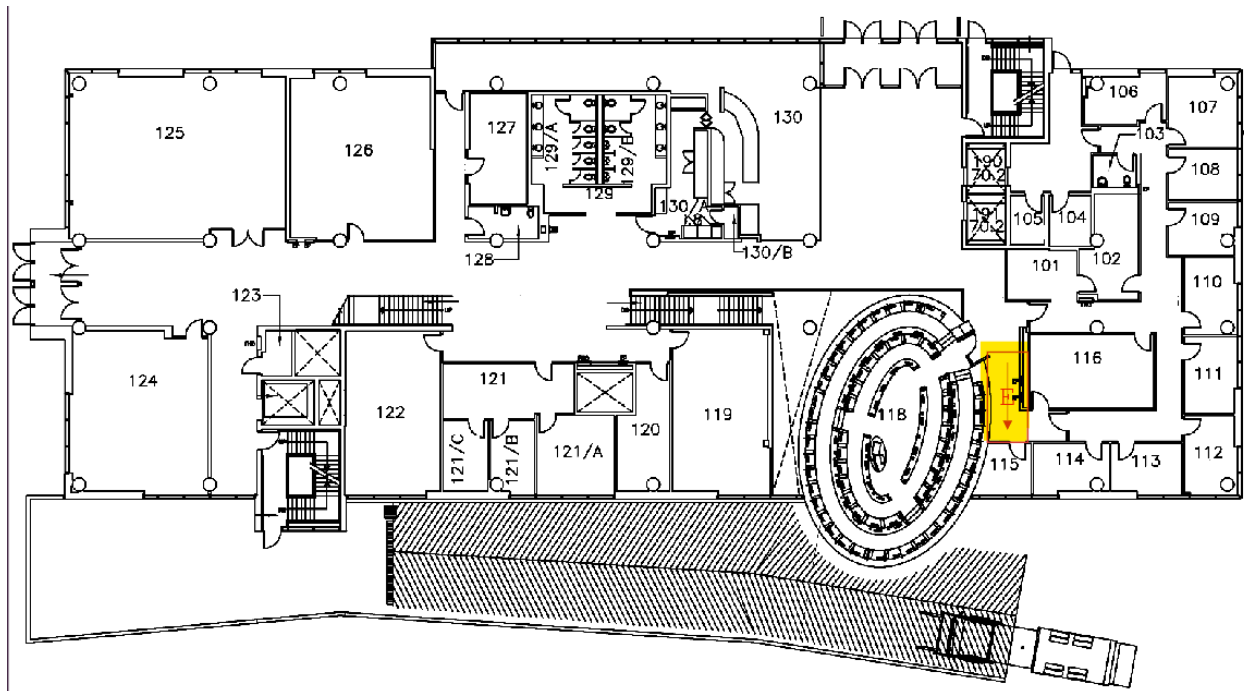


Figure 3. Engineering Technology Building (ETB) First Floor with Scanned Hallway Highlighted

Location Images

In the figure below, there are multiple images at different areas in the scanned hallway.

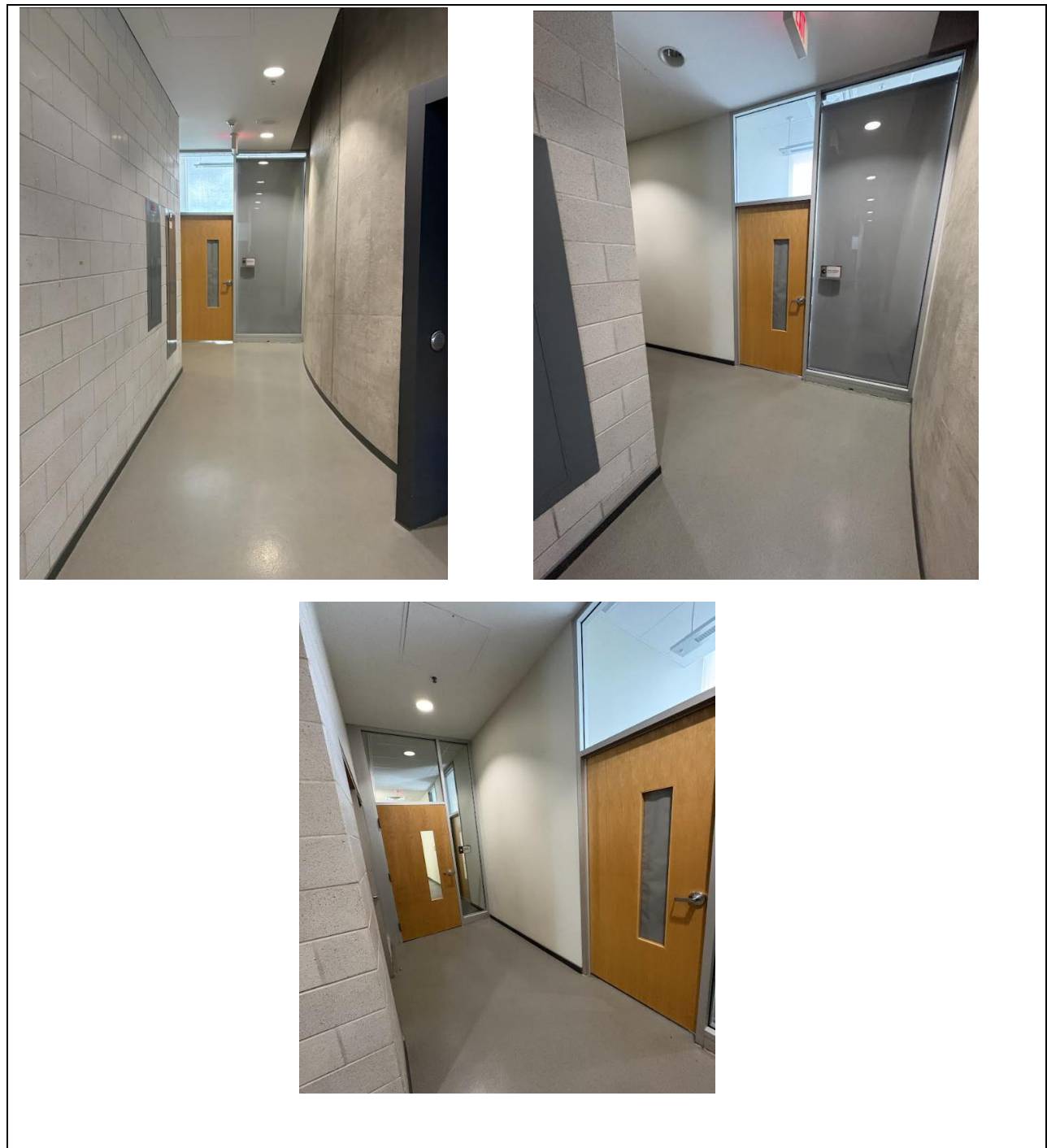


Figure 4. Images of Scanned Hallway

Expected Output

Below is the expected 3D mapping of the scanned hallway. Note that the stepper motor started on an angle, resulting in a slanted simulation.

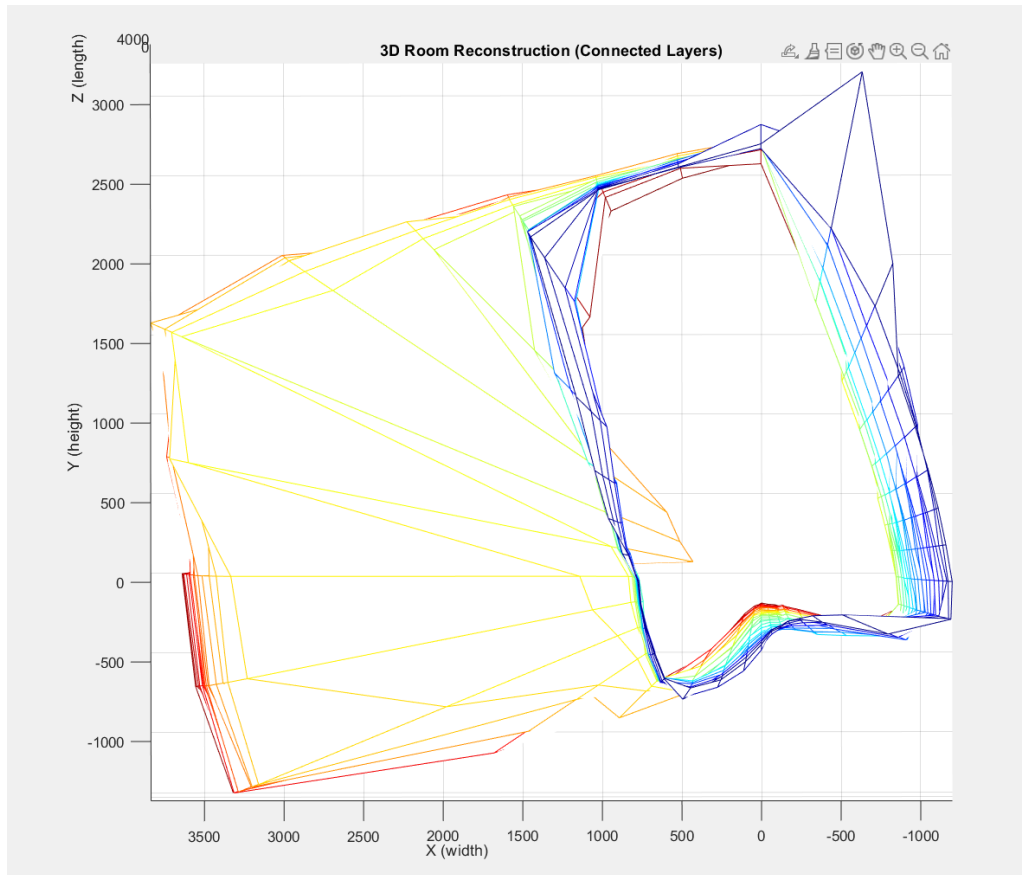


Figure 5. Front View of Scanned Hallway

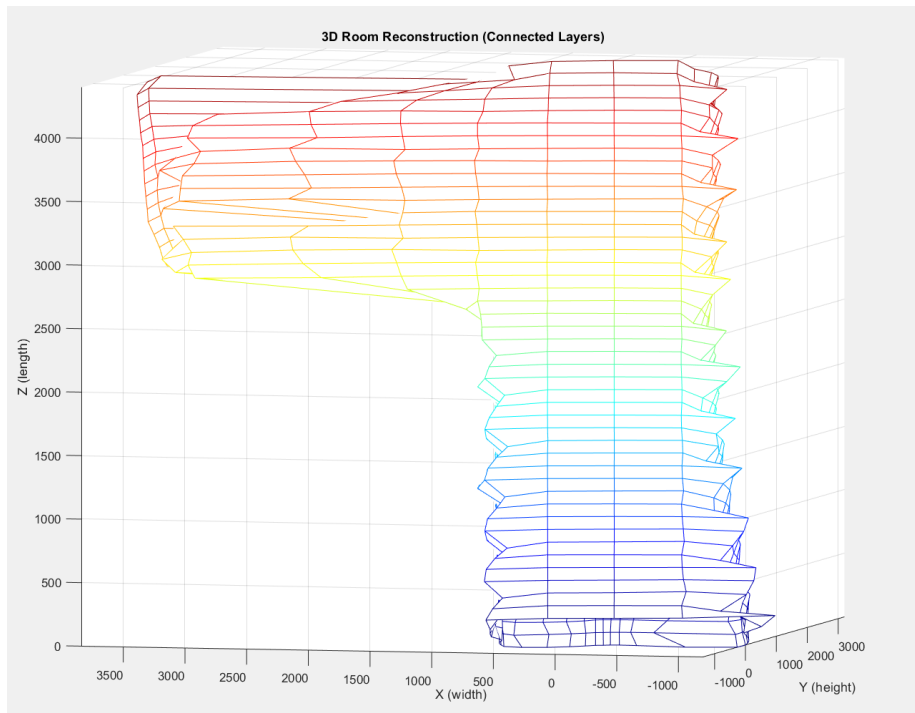


Figure 6. Top View of Scanned Hallway

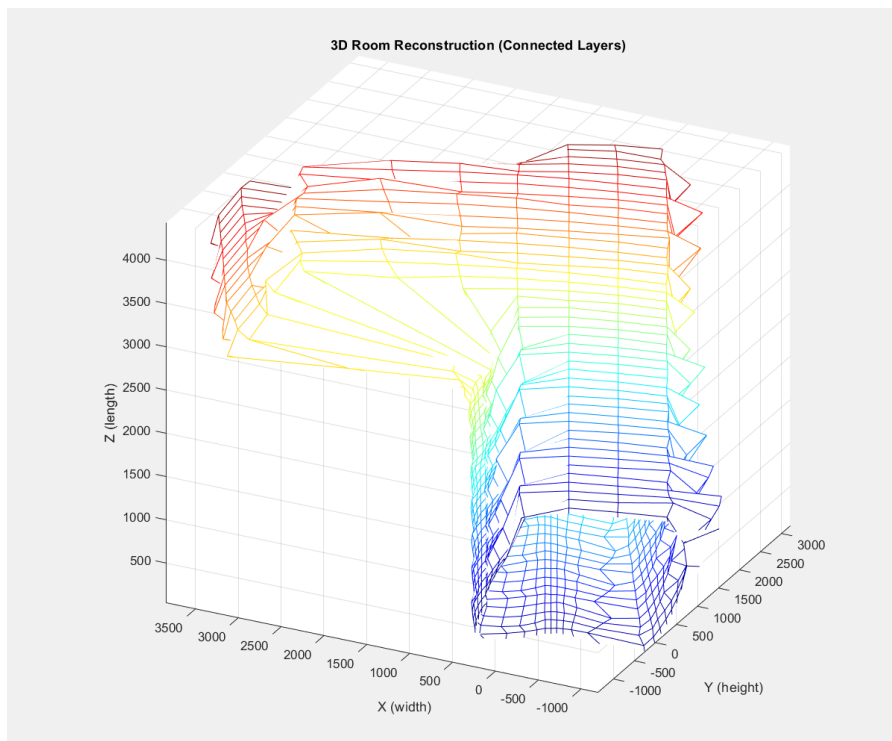


Figure 7. Bird's Eye View of Scanned Hallway

Limitations

While this system has its benefits to being a low cost, smaller LiDAR system, there are limitations to its capabilities.

Floating Point and Trigonometric Limitations

The MSP432E401Y features an integrated ARM Cortex-M4F core with a 32-bit Single-Precision Floating Point Unit (FPU). While this hardware can do basic math with high speed, it is limited to 32-bit float precision. In contrast, standard PC environments utilize 64-bit double precision, meaning it operates faster than the microcontroller.

The primary limitation arises when calculating trigonometric functions for spatial mapping. Using 32-bit precision for polar-to-Cartesian conversions results in rounding errors that are increased over large distances. Additionally, because the MCU must use MATLAB for 64-bit math, performing these calculations on-board would significantly increase execution time. To avoid this, the system sends the distance data to MATLAB, utilizing the PC's 64-bit FPU for coordinate transformation.

The maximum quantization error of the system can be calculated using the equation

$$\text{Max Quantization Error} = \frac{\text{Max Distance}}{2^n}$$

Where the max distance which the sensor can accurately measure is 4m and n, which is the resolution of the sensor data, is 16 bits. This gives a max quantization error of

$$\text{Max Quantization Error} = \frac{4m}{2^{16}}$$

$$\text{Max Quantization Error} = 0.061mm$$

Maximum Standard Serial Communication Rate

The maximum standard serial communication rate implemented between the microcontroller and the PC is 128,000 bits per second (bps).

This rate was verified by checking the 'Port Settings' of the XDS110 Class Application/User UART device within the Windows 11 Device Manager (COM 3), where 128,000 bps was the highest selectable option.

Communication Methods & Speed

Communication between the MSP432E401Y microcontroller and the VL53L1X Time-of-Flight (TOF) sensor is handled through the I2C. In this system, the microcontroller acts as the I2C leader, while the sensor acts as the follower.

The bus speed, which can reach a max speed of 400 kbps (Fast Mode), is configured to 100 kbps (Standard Mode).

Primary Limitation on Speed

The primary bottlenecks impacting the total system scan speed are the stepper motor and the TOF sensor.

1. **Stepper Motor Bottleneck:** The 28BYJ-48 stepper motor can only spin properly when a minimum time delay of 2ms is applied.
2. **Sensor Bottleneck:** The TOF sensor uses a timing budget (between 20ms and 1000ms) which impacts its accuracy and range of the distance measured. With a longer, the measurement is more accurate, but the scan duration is increased as a result. To ensure the stepper motor and TOF sensor are in sync, the stepper motor must pause between steps long enough for the TOF sensor to complete its measurement.

These limitations were verified through iterative testing. When decreasing the delay in between the motor's steps below 1.6ms, the motor vibrated without moving, confirming its limit. Similarly, reducing the TOF timing budget resulted in a significant increase in distance inaccuracies as well as range status errors for further measured distances.

Circuit Schematic

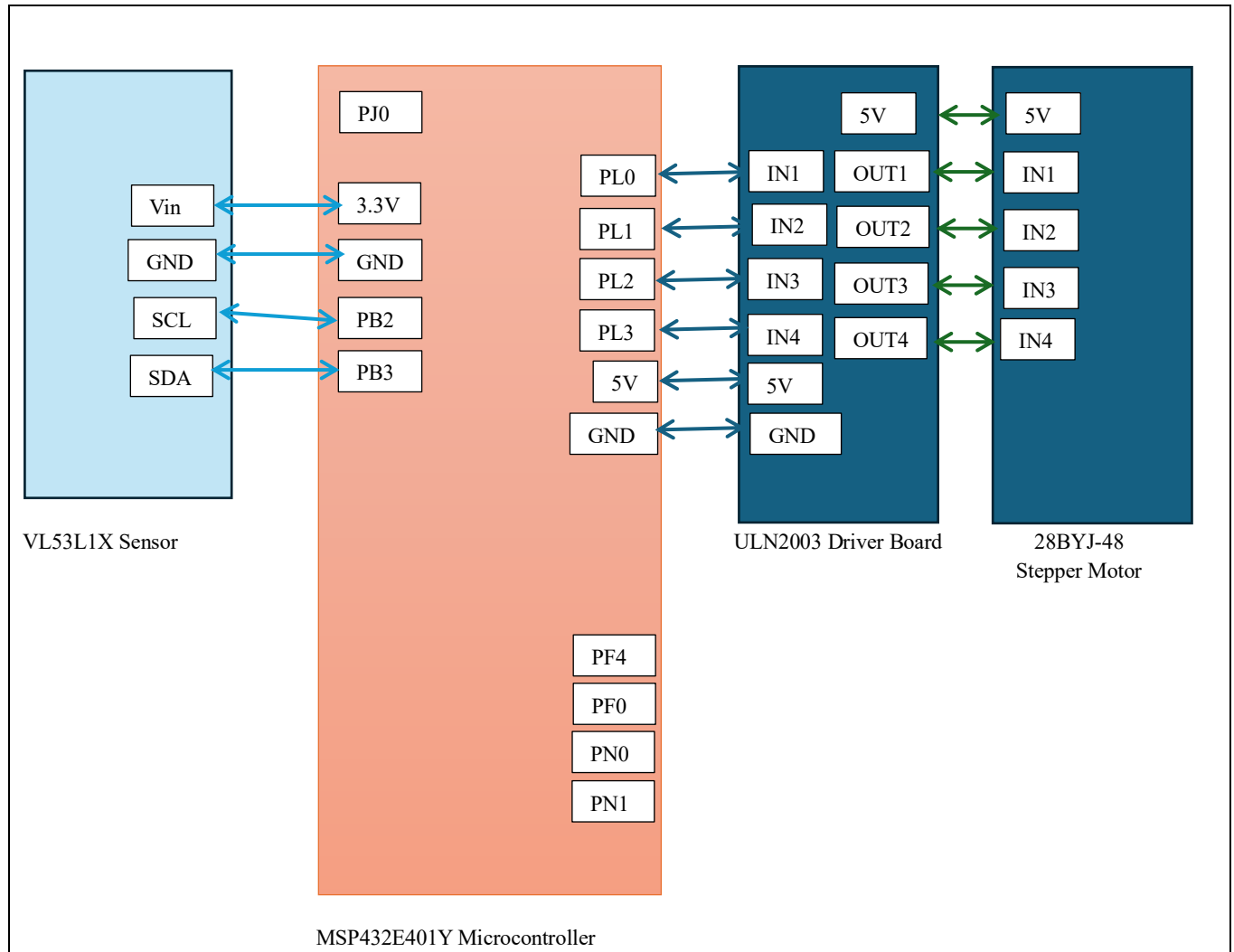


Figure 8. LiDAR System Circuit Schematic

Programming Logic Flowchart

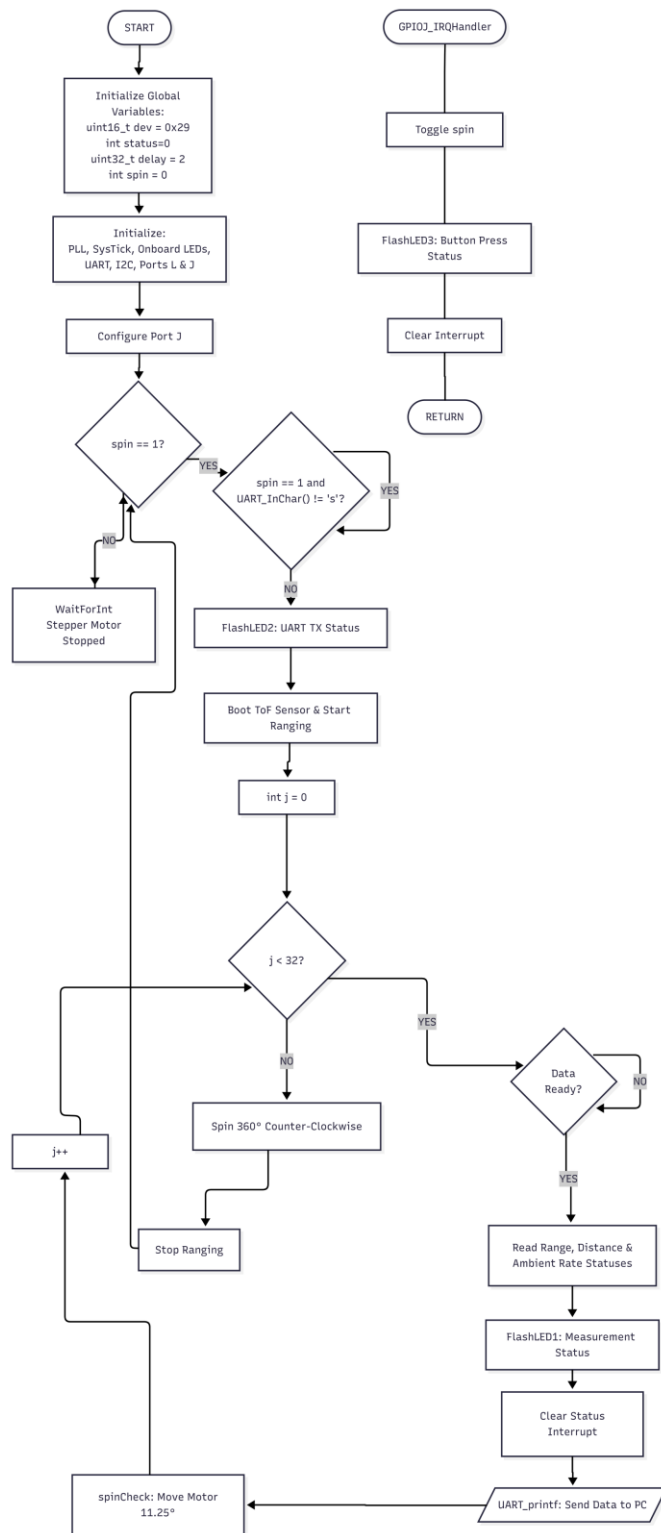


Figure 9. Keil Code Flowchart

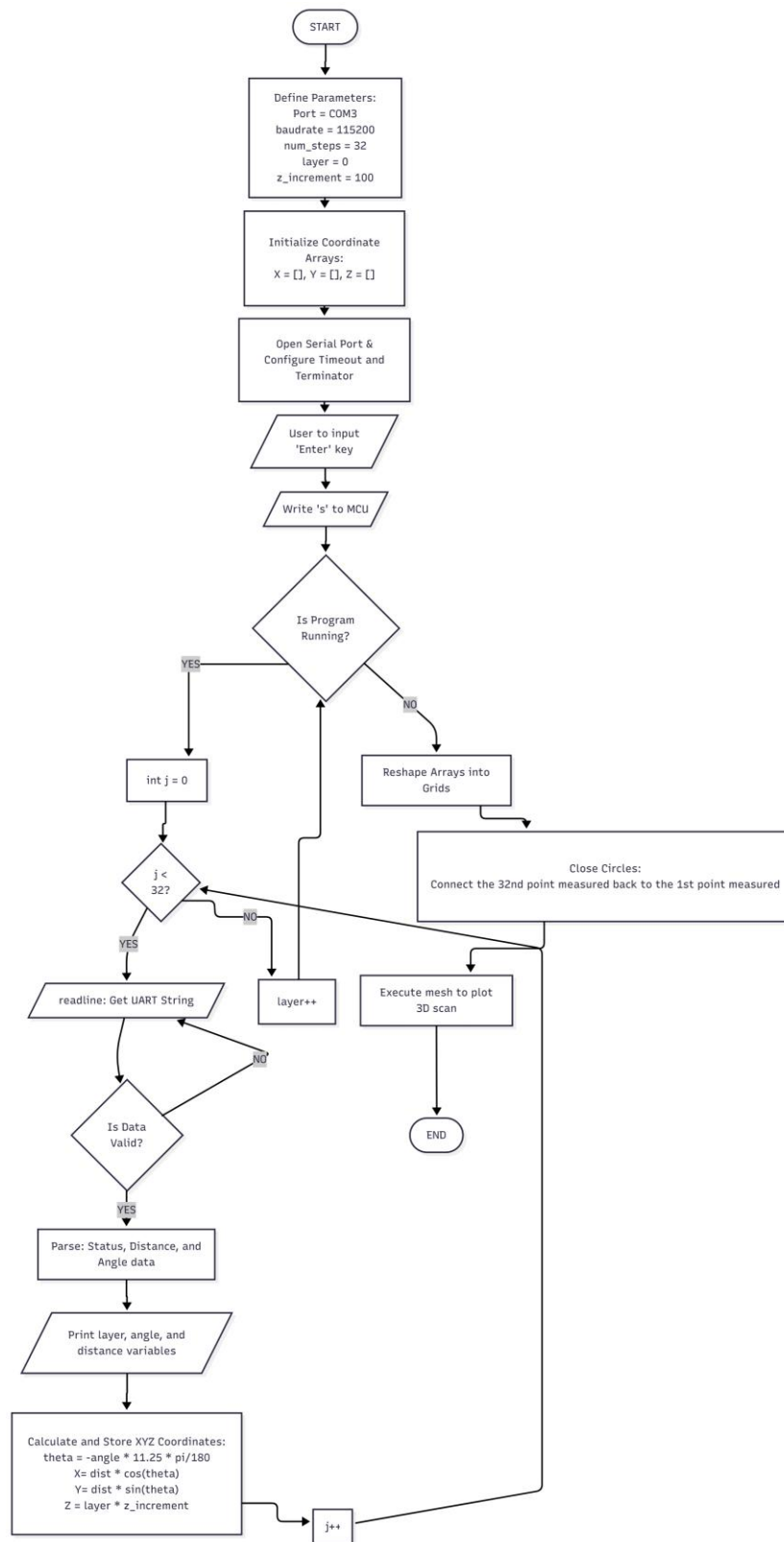


Figure 10. MATLAB Code Flowchart